Reinforcement learning algorithms solve Markov decision processes by estimating value functions or optimizing policies from sampled transitions. Two assumptions underlie the standard formulation. First, the agent's action at time t is determined solely by the observed state s_t and the agent's policy $\pi(a \mid s_t)$; there is no unobserved variable (confounder) simultaneously influencing both the action and the reward or state transition.[1] Second, the observed state s_t is sufficient for prediction, so that conditioning on s_t renders future states independent of past history. Both assumptions are the Markov property restated in causal language. Both fail routinely in economic applications.

The preceding chapters operated under a third assumption that was so natural it required no mention: the analyst controls data collection. In the tabular algorithms of Section **??** and the gridworld study of Section **??**, each method generated its own trajectories by executing actions in a simulator and observing the consequences. The bandit algorithms of Section **??** chose arms and observed payoffs in real time. Even when exploration was limited, the data-generating process was known because the agent's own policy produced it. This chapter drops that assumption entirely. The analyst receives a fixed log of decisions made by someone else, a *behavioral policy* μ whose functional form may be unknown and whose action choices may depend on variables the analyst cannot observe. The data are observational in the econometric sense: the analyst had no role in generating them and cannot rerun the experiment under a different policy. Identification, not optimization, becomes the central problem.

Unobserved demand shocks that affect both a retailer's pricing algorithm and consumer demand create endogeneity. The observed correlation between price and demand conflates the causal effect with the confounding effect. An RL agent trained on such observational data converges to a biased policy that systematically overestimates the revenue from high prices.

This chapter formalizes the *confounded MDP*, develops four identification strategies for recovering interventional quantities from observational data (backdoor adjustment, frontdoor adjustment, instrumental variables, and proximal causal inference), and demonstrates their practical consequences through a unified simulation study. For comprehensive surveys of the rapidly growing causal RL literature, including causal representation

[1] See Section **??** for the terminological mapping between "outcome" in causal inference and the corresponding RL quantities. Throughout this chapter, I reserve "outcome" for its causal inference meaning and use the specific RL quantity (reward, state, return, value) elsewhere.

learning, counterfactual policy optimization, transfer, and fairness, I refer readers to Deng et al. (2023) and da Costa Cunha et al. (2025).

0.1 From Partial Observability to Causal Structure

Before formalizing confounded MDPs, it is useful to distinguish partial observability from confounding, since both involve hidden variables but create fundamentally different challenges. A partially observable MDP (POMDP) augments the standard MDP with an observation function. The POMDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, r, \gamma)$, where $\mathcal{O}$ is a finite observation space and O is the observation function.

$$O(o \mid s', a) = P(O_t = o \mid S_t = s', A_{t-1} = a). \tag{1}$$

The agent does not observe s_t directly but instead receives $o_t \sim O(\cdot \mid s_t, a_{t-1})$ and maintains a belief state $b_t \in \Delta(\mathcal{S})$

$$b_t(s) = P(S_t = s \mid o_1, a_1, \ldots, o_t), \tag{2}$$

which is updated via Bayesian filtering at each step. The belief MDP, whose state space is $\Delta(\mathcal{S})$, is itself a fully observable (continuous-state) MDP, so standard value iteration applies in principle, though computation is intractable in general.

da Costa Cunha et al. (2025) organize sequential decision problems with hidden variables into a hierarchy: standard MDP (full observability, no confounding), POMDP (partial observability, no confounding), confounded MDP (full observability, confounding), and causal POMDP (both). The key distinction is epistemic versus identificational. In a POMDP, the hidden state is a modeling challenge: the agent acknowledges incomplete information and plans accordingly via the belief state, analogous to Kalman or Hamilton filtering in econometrics. In a confounded MDP, the hidden variable is an identification challenge: standard estimators silently produce biased results, analogous to endogeneity and omitted variable bias.

0.2 The Confounded MDP

When unobserved confounders influence both the behavioral policy and the transitions or rewards, the MDP is confounded. This formalization, developed by Zhang and Barein-

boim (2019), Zhang and Bareinboim (2020), and Kallus and Zhou (2020), provides the foundation for causal reasoning in sequential decision problems. The key tool is Pearl's do-operator. The interventional distribution $P(Y \mid \mathrm{do}(X = x))$ is the distribution that arises when X is set externally rather than observed passively, severing all incoming causal influences on X while leaving the remaining data-generating process intact.[2]

Definition D1 (Confounded MDP (Zhang and Bareinboim, 2020))**.** *A confounded MDP is a tuple* $(\mathcal{S}, \mathcal{A}, \mathcal{U}, P, r, \gamma)$ *where* $\mathcal{S}$ *is a finite state space,* $\mathcal{A}$ *is a finite action space,* $\mathcal{U}$ *is a space of unobserved confounders,* $\gamma \in [0, 1)$ *is a discount factor, and the dynamics are governed by structural equations*

$$U_t \sim P_U(\cdot \mid S_t), \tag{3}$$

$$A_t \sim \mu(\cdot \mid S_t, U_t), \tag{4}$$

$$R_t = f_R(S_t, A_t, U_t) + \epsilon_t, \tag{5}$$

$$S_{t+1} \sim f_S(\cdot \mid S_t, A_t, U_t). \tag{6}$$

The behavioral (logging) policy μ *depends on the unobserved confounder* U_t *through Equation (4). An evaluation policy* $\pi(a \mid s)$ *depends only on the observed state.*

Unlike the online algorithms of Sections **??**–**??**, where behavior and target policies were identical or related by a known exploration mechanism, here μ is an unknown function of unobserved variables.

Because μ depends on U_t, conditioning on $\{A_t = a\}$ carries information about the confounder, so the observational and interventional transitions diverge:

$$P(s' \mid s, a) \neq P(s' \mid s, \mathrm{do}(a)). \tag{7}$$

The Bellman equation for policy evaluation must use interventional, not observational, transition probabilities. Define the causal Bellman operator for a *target policy* π:

Definition D2 (Causal Bellman Operator (Zhang and Bareinboim, 2020))**.** *The causal*

[2]The do-operator is formalized within the structural causal model (SCM) framework of Pearl (2009). An SCM specifies endogenous variables $\mathbf{V}$, exogenous variables $\mathbf{U}$, structural equations $V_i = f_i(\mathrm{pa}(V_i), U_i)$, and a distribution $P(\mathbf{U})$. The intervention $\mathrm{do}(X = x)$ replaces the structural equation for X with a constant, producing the interventional distribution. See Pearl (2009) for the complete framework, including the causal hierarchy (association, intervention, counterfactual) and general identification theory.

Bellman operator $\mathcal{T}_c^\pi$ for policy π in a confounded MDP is

$$(\mathcal{T}_c^\pi V)(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, \mathrm{do}(a)) \Big[r(s,a) + \gamma V(s') \Big], \tag{8}$$

where $r(s,a) = \mathbb{E}[R_t \mid S_t = s, \mathrm{do}(A_t = a)]$ is the interventional expected reward.

Lemma L1 (Bias of Naive Off-Policy Evaluation (Kallus and Zhou, 2020))**.** *Let $\hat{V}^\pi_{naive}$ denote the value function obtained by solving the Bellman equation with observational transitions $P(s' \mid s, a)$, and let V^π denote the true value function under interventional transitions $P(s' \mid s, \mathrm{do}(a))$. In a confounded MDP where $P(s' \mid s, a) \neq P(s' \mid s, \mathrm{do}(a))$ for some (s, a, s'), the naive estimator is biased.*

$$\hat{V}^\pi_{naive}(s) \neq V^\pi(s). \tag{9}$$

The importance-sampling estimator $\hat{V}^\pi_{IS} = \frac{1}{N} \sum_{i=1}^{N} \prod_{t=0}^{T} \frac{\pi(a_t^{(i)} | s_t^{(i)})}{\mu(a_t^{(i)} | s_t^{(i)})} G^{(i)}$, where $G^{(i)} = \sum_{t=0}^{T} \gamma^t R_t^{(i)}$ is the discounted return of trajectory i and T is the trajectory length, is also biased because the propensity $\mu(a \mid s)$ is not the true behavioral propensity $\mu(a \mid s, u)$.[3]

The backdoor criterion (Pearl, 2009) provides a path to identification. Zhang and Bareinboim (2020) apply it to the confounded MDP setting.

Theorem 1 (Backdoor Identification in Confounded MDPs (Pearl, 2009; Zhang and Bareinboim, 2020))**.** *Suppose a set of observed variables $\mathbf{Z}_t$ satisfies the backdoor criterion relative to (A_t, S_{t+1}) in the causal graph of the confounded MDP, meaning that $\mathbf{Z}_t$ blocks all backdoor paths from A_t to S_{t+1} and no element of $\mathbf{Z}_t$ is a descendant of A_t.*[4] *Then the interventional transition probability is identified:*

$$P(s' \mid s, \mathrm{do}(a)) = \sum_{\mathbf{z}} P(s' \mid s, a, \mathbf{z})\, P(\mathbf{z} \mid s). \tag{10}$$

Substituting Equation (10) into the causal Bellman operator (Equation 8) yields an identified, consistent estimator of V^π.

[3] This is the sequential analogue of the omitted variable bias in linear regression. In the static case, regressing Y on X without controlling for a confounder U yields a biased coefficient. In the sequential case, the bias propagates through the Bellman recursion and can amplify over the horizon.

[4] A backdoor path from A_t to S_{t+1} is any path in the causal graph that begins with an arrow into A_t (i.e., a non-causal path). In the confounded MDP, $A_t \leftarrow U_t \rightarrow S_{t+1}$ is a backdoor path: U_t causes both A_t and S_{t+1}, creating a spurious association. Blocking all such paths by conditioning on appropriate variables eliminates the confounding bias. See Pearl (2009), Chapter 3.

0.3 Backdoor-Adjusted Off-Policy Evaluation

Off-policy evaluation under confounding is an average treatment effect estimation problem in a dynamic setting (Bannon et al., 2020): μ is the treatment assignment mechanism, π the counterfactual regime, importance sampling corresponds to inverse probability weighting, and doubly robust OPE corresponds to the AIPW estimator of Robins et al. (1994).

Theorem 1 yields a concrete estimation procedure. Given logged data $\{(s_t, a_t, z_t, r_t, s_{t+1})\}_{t=1}^{N}$ collected under behavioral policy μ, where z_t is an observed proxy for the confounder.

1. Estimate the conditional transition model $\hat{P}(s' \mid s, a, z)$ and the proxy distribution $\hat{P}(z \mid s)$ from the logged data.
2. Compute interventional transitions via the backdoor adjustment:

$$\hat{P}(s' \mid s, \mathrm{do}(a)) = \sum_{z} \hat{P}(s' \mid s, a, z)\, \hat{P}(z \mid s). \tag{11}$$

3. Solve the causal Bellman equation (Definition D2) using the estimated interventional transitions to obtain $\hat{V}^{\pi}$.

The doubly robust variant combines the fitted action-value function $\hat{Q}(s, a)$ (the RL analogue of the econometric outcome model) with backdoor-adjusted propensities, achieving consistency if either $\hat{Q}$ or the propensity model is correctly specified.

0.4 Alternative Identification Strategies

When no backdoor variable is available, three alternative identification strategies apply, each with a direct econometric analogue. Figure 1 displays the causal graph for each.

0.4.1 Front-Door Criterion

The front-door criterion applies when A_t affects S_{t+1} only through an observed mediator M_t. Three conditions are required: (i) M_t intercepts all directed paths from A_t to S_{t+1}, (ii) no unblocked backdoor path exists from A_t to M_t, and (iii) A_t blocks all backdoor paths from M_t to S_{t+1}. When satisfied (Pearl, 2009):

$$P(s' \mid s, \mathrm{do}(a)) = \sum_{m} P(m \mid a) \sum_{a'} P(s' \mid s, m, a')\, P(a' \mid s). \tag{12}$$

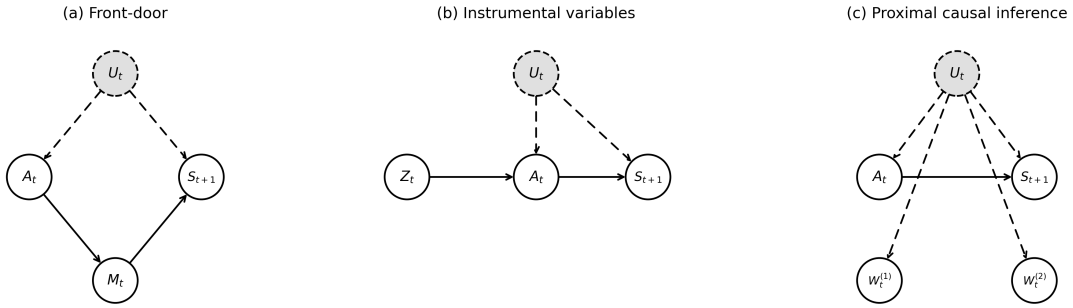

Figure 1: Causal graphs for three identification strategies in confounded MDPs. Gray dashed nodes are unobserved; dashed edges involve unobserved variables. (a) Front-door criterion with mediator M_t. (b) Instrumental variables with exogenous instrument Z_t. (c) Proximal causal inference with proxies $W_t^{(1)}, W_t^{(2)}$.

The first factor is unconfounded; the inner sum adjusts for confounding on the $M_t \to S_{t+1}$ link by averaging over the observational action distribution. This is the sequential analogue of mediation analysis (da Costa Cunha et al., 2025).

The causal RL survey of da Costa Cunha et al. (2025) illustrates the front-door criterion with a mobile wellness intervention. A health app (action A_t) aims to reduce patient cortisol levels (outcome S_{t+1}), but unobserved health consciousness U_t confounds the relationship because health-conscious individuals are both more likely to adopt the app and more likely to have low cortisol regardless. The app affects cortisol only through an observed mediator, supplement adherence M_t. Applying Equation (12) recovers the causal effect without observing health consciousness.

0.4.2 Instrumental Variables

When neither backdoor nor front-door variables are available, instrumental variable methods can identify causal effects. An instrument Z_t must satisfy two conditions: it affects the action A_t (relevance) and its effect on S_{t+1} is channeled entirely through A_t (exclusion restriction). Liao et al. (2024) formalize this as a Confounded MDP with Instrumental Variables (CMDP-IV), where transitions take the form $S_{t+1} = F^*(S_t, A_t) + \epsilon_t$ with unobserved confounders ϵ_t affecting both the behavior policy and transitions. The transition function F^* is recovered from a conditional moment restriction:

$$\mathbb{E}[S_{t+1} - F^*(S_t, A_t) \mid Z_t, S_t] = 0. \tag{13}$$

For a binary action and binary instrument, the Wald estimator provides a closed-form solution. Let β denote the causal effect of promoting (action $a = 0$) on the transition probability.

$$\beta = \frac{P(s' \mid s, Z{=}1) - P(s' \mid s, Z{=}0)}{P(A{=}0 \mid s, Z{=}1) - P(A{=}0 \mid s, Z{=}0)}, \tag{14}$$

where the numerator is the reduced-form effect of the instrument on the transition and the denominator is the first-stage effect on treatment uptake. The interventional transition is then recovered from any instrument value z via $P(s' \mid s, \mathrm{do}(a{=}0)) = P(s' \mid s, Z{=}z) + \beta \cdot (1 - P(A{=}0 \mid s, Z{=}z))$. Their IV-aided Value Iteration algorithm applies this moment restriction at each state to estimate F^*, then runs standard value iteration on the estimated model.

Liao et al. (2024) illustrate with neonatal intensive care unit (NICU) assignment. A hospital must decide whether to admit each newborn to the NICU (action A_t), and this decision is confounded by unobserved severity indicators that affect both the admission decision and the infant's health trajectory. Differential travel time from the hospital to a specialty care provider serves as an instrument Z_t. Relevance holds because longer travel times discourage NICU referrals, shifting the admission probability. The exclusion restriction holds because travel time affects infant health outcomes only through the admission decision, not directly. The conditional moment restriction (Equation 13) uses variation in travel time across hospitals to trace out the causal effect of NICU admission on health transitions, recovering the transition function F^* that standard regression conflates with the unobserved confounder.

0.4.3 Proximal Causal Inference

When confounders are truly latent but proxy variables, noisy correlates of the confounder, are available, proximal causal inference provides identification. Bennett and Kallus (2021) adapt this to sequential settings. The analyst observes two proxies: a treatment-side proxy $W_t^{(1)}$ and an outcome-side proxy $W_t^{(2)}$, both conditionally independent given the latent confounder U_t. Identification proceeds through a bridge function h that solves a conditional moment equation linking the two proxies to the interventional quantity:

$$\mathbb{E}[V(S_{t+1}) \mid W_t^{(1)}, S_t, A_t] = \mathbb{E}[h(W_t^{(2)}, S_t, A_t) \mid W_t^{(1)}, S_t, A_t]. \tag{15}$$

The bridge function h is estimated by solving this integral equation (which reduces to a linear system in the discrete case), and the causal effect is recovered by marginalizing over the outcome proxy distribution.

$$P(s' \mid s, \mathrm{do}(a)) = \sum_{w_2} h(w_2, s, a)\, P(W^{(2)}{=}w_2 \mid s). \tag{16}$$

Two bridge functions, analogous to inverse propensity scores and Q-functions, yield a doubly robust estimator of the policy value that is $\sqrt{n}$-consistent without ever observing U_t. Bennett and Kallus (2021) demonstrate with a sepsis management simulator in which physicians choose among fluids, vasopressors, and antibiotics at each decision point. The patient's diabetes status is the latent confounder, partially censored in the medical record so that 20% of diabetic patients appear as non-diabetic in the data. Because the physician observes the true diabetes status when prescribing but the analyst's dataset contains the censored version, the behavioral policy depends on a variable the analyst cannot fully recover. Previous clinical observations O_{t-1} (prior lab values and vitals) serve as the treatment-side proxy $W_t^{(1)}$ because they correlate with diabetes status and influence the physician's current prescribing. Current clinical observations O_t serve as the outcome-side proxy $W_t^{(2)}$ because they reflect diabetes status and predict future health transitions. In their experiments, the proximal estimator correctly identified which evaluation policy improved over the behavioral policy in 82–100% of test cases, while naive estimators that ignored confounding and standard MDP estimators that assumed full observability both achieved 0% accuracy.

0.5 The Broader Causal RL Landscape

Three active research directions beyond confounded MDPs illustrate the broader scope of causal RL.

Causal representation learning for RL seeks state representations $\phi(s)$ that capture causal mechanisms rather than spurious correlations (Schölkopf et al., 2021). da Costa Cunha et al. (2025) formalize this as invariant policy optimization within a multi-environment MDP framework. In sim-to-real robotics, training across multiple visually distinct simulators (same physics, different rendering) forces the representation to discard renderer-specific features and retain only causally relevant ones like object pose and joint angles.

Counterfactual policy optimization uses structural causal models to generate "what-if" trajectories for credit assignment and data augmentation. Buesing et al. (2019) introduce Gumbel-Max SCMs that produce counterfactual rollouts from a single observed trajectory via Pearl's Abduction-Action-Prediction procedure (Pearl, 2009): infer the exogenous noise explaining the observed trajectory, replace the logged action, and propagate through the structural equations. By replaying a trajectory with one action changed while holding the environment's randomness fixed, any difference in reward is attributable to that action, solving long-horizon credit assignment without importance sampling. Oberst and Sontag (2019) extend this to healthcare settings, while Forney et al. (2017) use counterfactual data-fusion to augment online exploration.

Causal transfer in RL applies the transportability theory of Pearl and Bareinboim (2014) and Bareinboim and Pearl (2016) to policy transfer across domains. Selection diagrams identify which mechanisms are shared across environments and which differ, enabling targeted recalibration rather than wholesale domain adaptation. In sim-to-real autonomous driving, physical dynamics transfer directly while visual rendering requires recalibration with scarce real-world data (Bareinboim and Pearl, 2016).

0.6 Simulation Study: Confounded Retail Pricing MDP

I construct a 5-state engagement funnel $\mathcal{S} = \{0, 1, 2, 3, 4\}$ with two actions (promote, hold price) and an absorbing conversion state at $s = 4$. A retailer manages customers through engagement stages, deciding whether to offer a promotional discount. The data-generating process embeds four distinct sources of causal variation, enabling simultaneous validation of all four identification strategies from a single DGP.

Figure 2 displays the complete causal graph. Market conditions $Z_t \sim \text{Bernoulli}(0.5)$ are observed and affect both the latent confounder and transitions. Consumer sentiment U_t is an unobserved confounder strongly correlated with Z_t.[5] An independent cost shock $\text{IV}_t \sim \text{Bernoulli}(0.5)$ serves as an instrument. The behavioral pricing policy depends on both U_t and IV_t: $\mu(\text{promote} \mid s, U_t, \text{IV}_t) = 0.55 + \rho \cdot 0.25 \cdot (2U_t - 1) + 0.15 \cdot (\text{IV}_t - 0.5)$, where $\rho \in \{0, 0.2, \ldots, 1.0\}$ controls confounding strength. Promotions trigger marketing follow-ups with M_t serving as a mediator.[6] Two noisy proxies of U_t are available: a CRM

[5] $P(U_t{=}1 \mid Z_t{=}1) = 0.9$ and $P(U_t{=}1 \mid Z_t{=}0) = 0.1$.

[6] $M_t \sim \text{Bernoulli}(0.8)$ when the retailer promotes and $\text{Bernoulli}(0.2)$ otherwise.

score $W_t^{(1)}$ and browsing behavior $W_t^{(2)}$.[7]

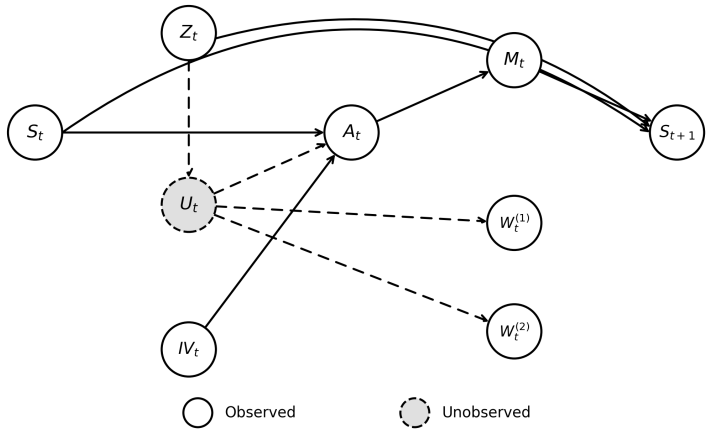

Figure 2: Complete causal graph of the simulation DGP. Gray dashed node (U_t) is unobserved; dashed edges involve U_t.

The action affects the next state only through the mediator: $P(s{+}1 \mid s, M_t, Z_t)$ depends on M_t and Z_t but not on A_t or U_t directly. This enables all four identification strategies simultaneously: Z_t satisfies the backdoor criterion, M_t satisfies the front-door criterion, IV_t satisfies relevance and exclusion, and $W_t^{(1)}, W_t^{(2)}$ satisfy the proximal conditions.[8]

I compare six estimators: oracle, naive (biased per Lemma L1), backdoor (Equation 11), front-door (Equation 12), Wald IV (Equation 14), and proximal (Equations 15–16).[9]

Figure 3 reports bias and RMSE across confounding strengths (20 seeds, 2,000 trajectories per seed). The naive estimator's bias grows monotonically with ρ because observational transitions overestimate promotion success: the promote action is more likely when $U_t = 1$, which correlates with favorable conditions, so $\hat{P}_{\text{obs}}$ exceeds the true interventional probability, and this per-step bias compounds through the Bellman recursion. The backdoor and front-door estimators eliminate bias at all ρ, validating Theorem 1 and

[7] $W_t^{(1)} \sim \text{Bernoulli}(0.85 \cdot U_t + 0.15 \cdot (1 - U_t))$ and $W_t^{(2)} \sim \text{Bernoulli}(0.75 \cdot U_t + 0.25 \cdot (1 - U_t))$.

[8] Rewards are $r(s, a) = -1$ for $s < 4$, $\gamma = 0.9$. The target policy always promotes. The true interventional transition probability is $P(s{+}1 \mid s, \text{do}(\text{promote})) = 0.615$.

[9] Each configuration uses 2,000 trajectories averaged over 20 seeds.

Equation (12). The IV estimator maintains low bias but higher variance due to the Wald ratio's sensitivity to instrument strength; panel (c) illustrates this classic bias-variance tradeoff. The proximal estimator achieves low bias with moderate variance, confirming that bridge functions recover causal effects from noisy proxies.

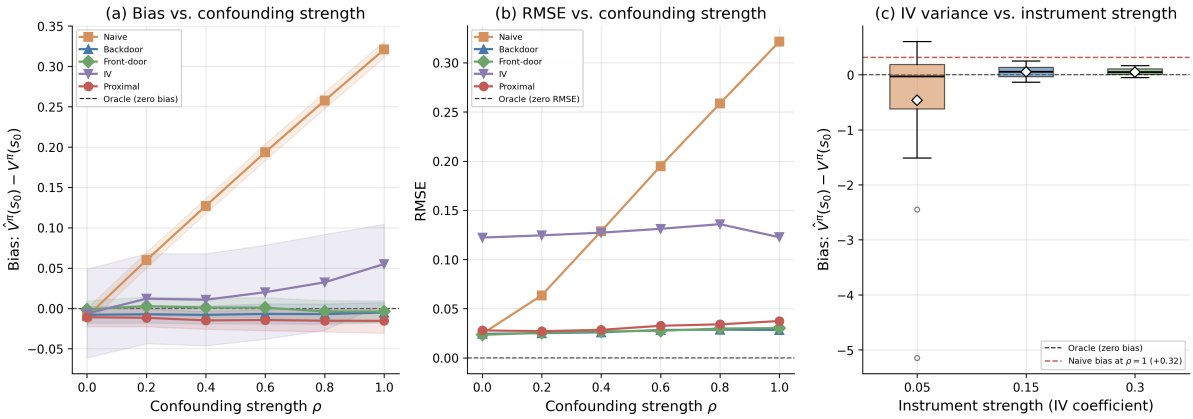

Figure 3: (a) Bias of five OPE estimators as a function of confounding strength ρ. (b) RMSE of five estimators as a function of ρ. (c) IV estimator bias distribution vs. instrument strength at $\rho = 1$; dashed red line is naive estimator bias.

References

James Bannon, Brad Langlois, Raimundo Fernandez, and Danielle Maddix. Causality and batch reinforcement learning: Complementary approaches to planning in unknown domains. In *NeurIPS Workshop on Causal Discovery and Causality-Inspired Machine Learning*, 2020.

Elias Bareinboim and Judea Pearl. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352, 2016.

Andrew Bennett and Nathan Kallus. Proximal reinforcement learning: Efficient off-policy evaluation in partially observed Markov decision processes. *Operations Research*, 2021.

Lars Buesing, Theophane Weber, Yori Zwols, Sebastien Racaniere, Arthur Guez, Jean-Baptiste Lespiau, and Nicolas Heess. Woulda, coulda, shoulda: Counterfactually-guided policy search. In *International Conference on Learning Representations*, 2019.

Cristiano da Costa Cunha, Wei Liu, Tim French, and Ajmal Mian. Unifying causal reinforcement learning: Survey, taxonomy, algorithms and applications. *arXiv preprint arXiv:2512.18135*, 2025.

Zhihong Deng, Jing Jiang, Guodong Long, and Chengqi Zhang. Causal reinforcement learning: A survey. *Transactions on Machine Learning Research*, 2023.

Andrew Forney, Judea Pearl, and Elias Bareinboim. Counterfactual data-fusion for online reinforcement learners. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1156–1164, 2017.

Nathan Kallus and Angela Zhou. Confounding-robust policy evaluation in infinite-horizon reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Luofeng Liao, Zuyue Fu, Zhuoran Yang, Yixin Wang, Dingli Ma, Mladen Kolar, and Zhaoran Wang. Instrumental variable value iteration for causal offline reinforcement learning. *Journal of Machine Learning Research*, 25(303):1–56, 2024.

Michael Oberst and David Sontag. Counterfactual off-policy evaluation with Gumbel-Max structural causal models. In *Proceedings of the 36th International Conference on Machine Learning*, pages 4923–4932, 2019.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition, 2009.

Judea Pearl and Elias Bareinboim. External validity: From do-calculus to transportability across populations. *Statistical Science*, 29(4):579–595, 2014.

James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.

Junzhe Zhang and Elias Bareinboim. Near-optimal reinforcement learning in dynamic treatment regimes. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Junzhe Zhang and Elias Bareinboim. Designing optimal dynamic treatment regimes: A causal reinforcement learning approach. In *Proceedings of the 37th International Conference on Machine Learning*, pages 11012–11022, 2020.